NeST-Com: Fixed-Budget Next-Scale Token Communication

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Abstract

Complete-scale endpoints can leave unused capacity in fixed-budget image communication. NeST-Com combines within-scale token endpoints, fitting by actual terminated entropy-code length, and prefix-constrained reconstruction with a shared frozen next-scale prior. On 500 ImageNet images, within-scale transmission improves PSNR by 0.32–0.70 dB over complete-scale transmission with the same model components; source-bit utilization at 10 dB rises from 77.70% to 95.21%. Controlled endpoint diagnostics and Kodak evaluation support this mechanism. A six-system comparison shows higher PSNR than a fixed-frame Ada-TokenCom adaptation, while Ada achieves better perceptual or feature-similarity scores at several operating points.

Index Terms

Image communication, visual autoregressive models, fixed symbol budget, entropy coding.

I. Introduction

An image communication system with a hard per-image symbol budget must decide which information to retain before channel transmission. The decision depends on the image-dependent source-code length, channel protection, and the reconstruction attainable from the retained representation. A generative receiver can supply missing information, but its output must remain

constrained by the information actually delivered. The resulting design trades pixel fidelity, perceptual similarity, feature consistency, and processing cost.

Digital image coding with forward error correction and learned joint source–channel coding (JSCC) offer different approaches to this tradeoff [1], [2]. Progressive transmission and truncation of embedded descriptions have a long history [3]. Recent token-based methods use large generative models: Ada-TokenCom combines prefix entropy coding, tail generation, and joint prefix/MCS selection under an average resource constraint [4]; TONIC studies task-oriented token communication [5]. ARPC constructs progressive image descriptions with an autoregressive prior [6]. Visual autoregressive modeling (VAR) provides a multiscale residual representation and a prior that predicts each scale from complete coarser maps [7].

For this next-scale representation, restricting transmission to complete maps gives large endpoint steps. Meanwhile, the terminated entropy-code length of a given endpoint varies across images. A next whole scale may exceed the frame even when the preceding whole endpoint leaves substantial room. Legal within-scale endpoints address this mismatch by retaining part of the boundary map. Their benefit concerns endpoint granularity within the same representation, rather than a finer truncation rule than Ada's individual-token prefixes.

We present NeST-Com, which connects within-scale endpoints, actual-length fitting, and prefix-constrained reconstruction in one fixed-budget communication design. The receiver uses the existing known-token overwrite operation of VAR while updating both the accumulated residual latent and the context for finer scales. Our contributions are the endpoint/frame-fitting design and its evaluation through shared-model comparisons, controlled granularity diagnostics, and six-system experiments. On common500, quarter-scale endpoints improve PSNR by 0.32–0.70 dB over complete-scale endpoints. External comparisons expose complementary pixel and perceptual behavior rather than a uniform ranking.

II. Proposed NeST-Com Framework

A. Next-Scale Representation and Digital Frame

We transmit $x \in [0,1]^{256\times256\times3}$ over complex AWGN,

$$y = s + n, \quad n \sim \mathcal{CN}(0, \sigma^2 I_N), \quad \mathrm{SNR} = 10\log_{10}(E_s/\sigma^2), \tag{1}$$

with $N = 1024$ complex symbols per image and nominal $E_s = 2$. The channel-use ratio to RGB samples is $1/192$. A 12-bit profile ID, CRC16, and terminated rate-1/2 convolutional code occupy 68 rotated-BPSK symbols $\{\pm(1+j)\}$. The remaining 956 symbols carry the LDPC-coded body. The paid profile identifies the actual endpoint and MCS.

The frozen VAR tokenizer produces ten residual token maps q_s, with side lengths $(1, 2, 3, 4, 5, 6, 8, 10, 13, 16)$ and a 4096-entry codebook [7]. Let L_s be the map size and C_s its codebook lookup, resizing, and residual transform. The complete quantized latent is $f_q = \sum_s C_s(q_s)$. Image decoder D_c is adapted on 20 000 ImageNet training images using continuous, fully quantized, and prefix-completion-interpolated latents, then fixed. NeST-Com uses null-class conditioning. Its frozen next-scale prior factorizes as

$$p_\theta(q_s \mid q_{<s}) = \prod_{i=1}^{L_s} p_\theta(q_{s,i} \mid q_{<s}), \tag{2}$$

providing the local cumulative distribution functions (CDFs) for arithmetic coding [8] at both ends of Fig. 1.

B. Within-Scale Endpoints and Actual-Length Fitting

An endpoint $e = (m, K)$ retains complete coarse maps and K raster-ordered positions from the next map:

$$\pi_e = (q_1, \ldots, q_m, q_{m+1,1:K}). \tag{3}$$

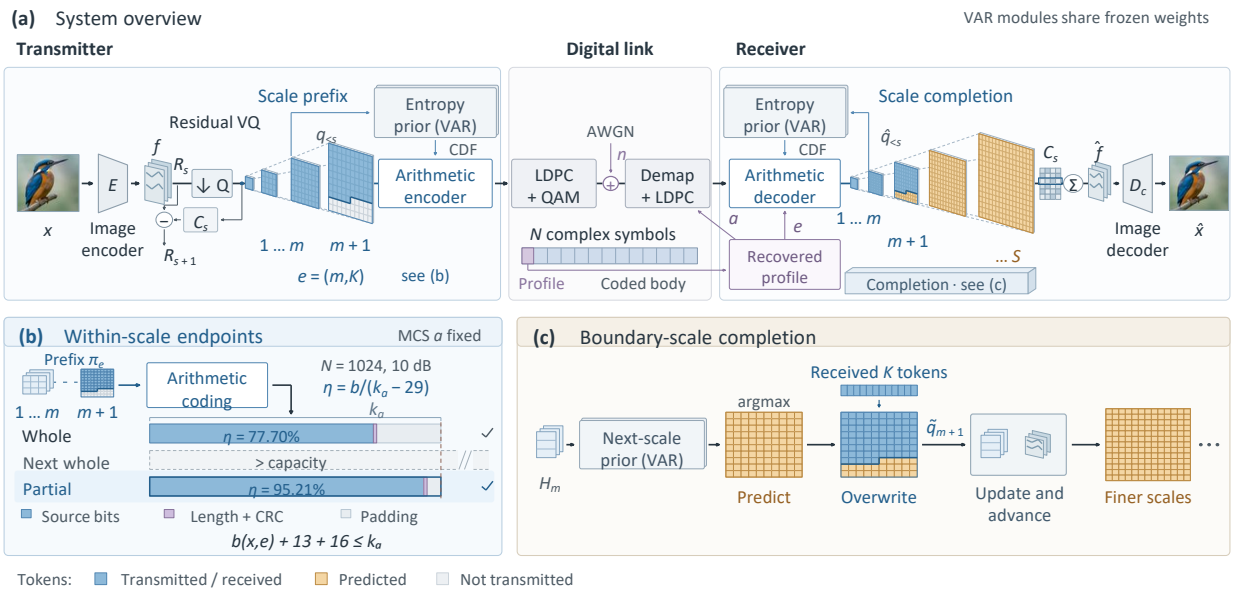


Fig. 1. NeST-Com. (a) A paid profile specifies endpoint e and MCS a; TX/RX construct CDFs locally using the shared frozen VAR and their known coarse tokens. (b) Independently terminated prefixes, a 13-bit length field, and CRC16 fit the information capacity k_a. Bars illustrate common500 mean utilization at 10 dB, not an individual bitstream. (c) Received positions overwrite the predicted boundary map before residual accumulation and advancement to finer scales. H_m denotes the prior state after scale m.

WHOLE permits $K = 0$. PARTIAL additionally uses $K \in \{\lfloor L_{m+1}/4 \rfloor, \lfloor L_{m+1}/2 \rfloor, \lfloor 3L_{m+1}/4 \rfloor\}$ for $m = 4, \ldots, 9$; $(10, 0)$ denotes the full representation.

For MCS $a = (M, r)$, the coded body has $n_a = 956 \log_2 M$ bits and information length $k_a = \lfloor r n_a \rfloor$. With actual terminated arithmetic length $b(x, e)$,

$$b(x, e) + 29 \le k_a, \qquad \eta(x, e) = \frac{b(x, e)}{k_a - 29}. \tag{4}$$

The 29 bits comprise the length field and CRC16; unused information positions are zero-padded before 5G NR LDPC coding, rate matching [9], and QAM. Thus η measures entropy-stream capacity utilization.

Offline calibration selects an SNR-specific target endpoint, MCS, and ordered fallback chain

using mean source/reconstruction DINOv2 cosine similarity [10]. Online, the transmitter checks the target's terminated length and moves along that chain until (4) holds, keeping the MCS fixed. The profile records the endpoint actually sent. This separates quality-based policy selection from per-image hard-budget fitting; an entropy estimate or a cut portion of a longer arithmetic stream is not the transmitted length.

C. Prefix-Constrained Reconstruction

Invalid profile CRC/catalogue, body CRC, declared length, known padding, or arithmetic parsing produces a fixed gray image. Otherwise, the receiver decodes the prefix using CDFs rebuilt from the complete coarse maps actually received. It predicts the boundary map by VAR argmax, then applies known-token overwrite [7]:

$$\tilde{q}_{m+1,i} = \begin{cases} \hat{q}_{m+1,i}, & i \le K, \\ q^{\mathrm{pred}}_{m+1,i}, & i > K. \end{cases} \tag{5}$$

The merged map contributes directly to the residual latent and updates the prior state before finer-scale prediction. Complete received maps remain unchanged; subsequent maps are generated from this updated context. Reconstruction is

$$\hat{x} = D_c\left(\sum_{s=1}^{10} C_s(\tilde{q}_s)\right). \tag{6}$$

At the final scale, overwrite changes the residual contribution without any subsequent propagation.

III. Experimental Results

A. Experimental Setup

We use common500 ImageNet validation images [11], a disjoint 1000-image calibration set, and the 24 Kodak images [12], with fixed 256×256 preprocessing. Tests use $N = 1024$, $E_s = 2$,

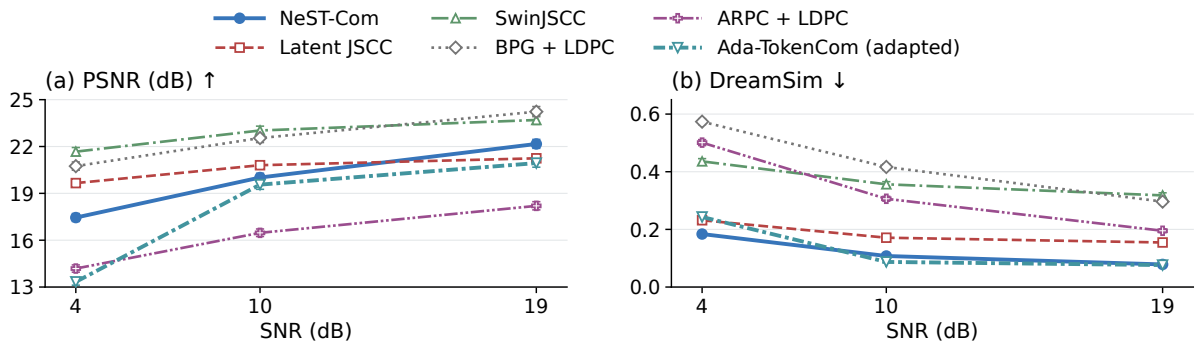


Fig. 2. Six-system performance on common500 at $N = 1024$. Points and bars are source means and pointwise 95% source-bootstrap CIs after averaging three channel realizations. Ada denotes the predicted-class, fixed-frame adaptation. Higher PSNR and lower DreamSim are favorable.

and 4/10/19 dB. ImageNet-calibrated policies remain fixed on Kodak.

The six systems are NeST-Com, Latent JSCC, adapted SwinJSCC [2], adaptive BPG [13] + LDPC, adapted ARPC [6] + LDPC, and Ada-TokenCom adapted to a fixed per-image frame. Our convolutional Latent JSCC maps tokenizer latents to symbols and shares D_c; its losses combine image MSE, LPIPS, and normalized-latent consistency. Latent JSCC and SwinJSCC use the same 20 000 training images and final 100k/200k-update checkpoints. SwinJSCC was still improving at its training budget cap. All required signaling is charged; ARPC's caption is paid. Ada retains LlamaGen-L/VQ16, prefix arithmetic coding, and conditional tail sampling. A fixed ConvNeXt predicts the class, whose 10-bit ID is transmitted, giving $b+39 \leq k_a$. Calibration chooses its MCS; online fitting searches the predeclared 1–256-token endpoints. This changes Ada's average-resource formulation to our fixed-frame task.

Metrics are PSNR, LPIPS-Alex [14], DreamSim [15], and CLIP-I, the CLIP ViT-L/14 image–image cosine [16]. DreamSim/CLIP use frozen FP32, 224×224 preprocessing. We average three channel realizations per source, retaining gray failures, then compute equal-source means.

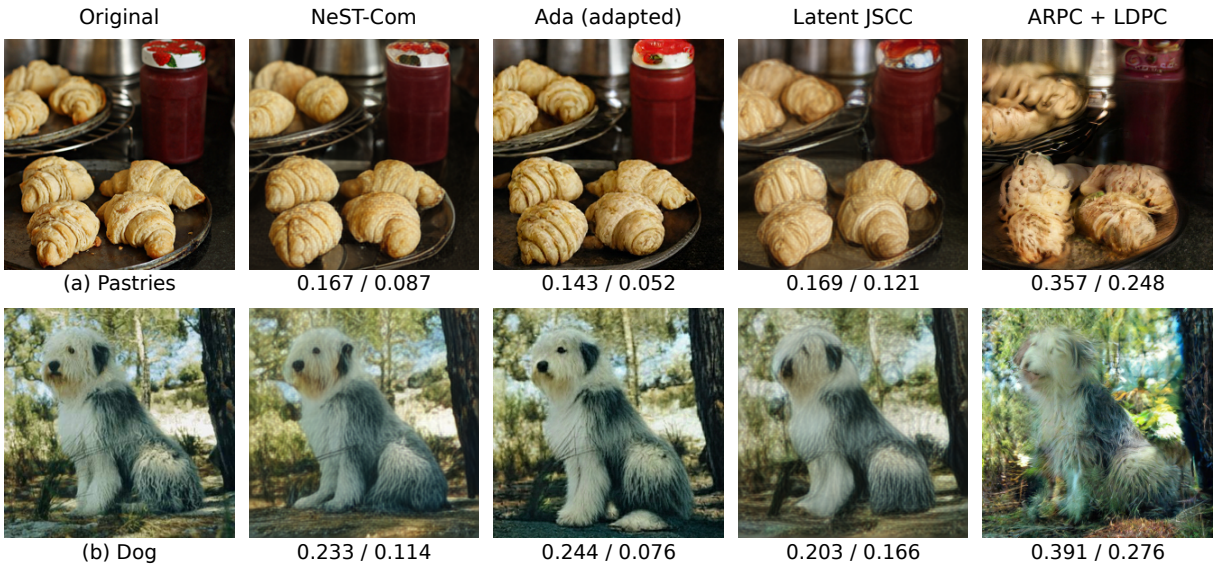


Fig. 3. Selected methods on two fixed development examples at $N = 1024$ and 10 dB: (a) pastries and (b) a dog. Values below each reconstruction are LPIPS-Alex↓ / DreamSim↓ for that displayed realization.

Intervals are 95% source-bootstrap CIs; differences are directly source-paired. Pairing does not require identical noise realizations across methods. Generative seeds and decoding settings are fixed.

B. Effect of Within-Scale Endpoints

Table I(A) compares WHOLE/PARTIAL with shared tokenizer, prior, entropy coder, and D_c. Their target/fallback policies are calibrated separately and select the same MCS at each SNR. On common500, PSNR gains are 0.32–0.70 dB. At 10 dB, utilization rises from 77.70% to 95.21% and DreamSim decreases by 0.0240. Kodak gains 0.674 dB and reduces DreamSim by 0.0164 at 10 dB; its 19 dB DreamSim CI includes zero. These results connect additional legal endpoints and calibrated fitting to better use of a fixed body capacity.

Block B isolates granularity on 100 existing sources, holding model, MCS, maximum endpoint,

TABLE I

WITHIN-SCALE ENDPOINT EVIDENCE. (A) MAIN FAMILY-SPECIFIC POLICIES; $\Delta = \mathrm{PARTIAL} - \mathrm{WHOLE}$ WITH SOURCE-PAIRED 95% CIS. (B) CONTROLLED 100-SOURCE DIAGNOSTIC AT 10 DB.

(A) Main policies: utilization and paired PSNR gain

SNR	Utilization W→P (%)	ΔPSNR (dB) [95% CI]
common500		
4	73.81 → 93.08	$+0.616\ [0.564, 0.673]$
10	77.70 → 95.21	$+0.701\ [0.637, 0.766]$
19	85.24 → 94.73	$+0.319\ [0.290, 0.349]$
Kodak24		
4	75.69 → 93.43	$+0.396\ [0.236, 0.543]$
10	78.11 → 94.99	$+0.674\ [0.408, 0.958]$
19	85.69 → 93.71	$+0.228\ [0.140, 0.324]$

(B) Fixed model, MCS, maximum endpoint and fitting

Grid (candidates)	Utilization (%)	PSNR (dB)
Whole (7)	78.15	18.952
Half (13)	89.21	19.396
Quarter (25)	95.38	19.612
Eighth (49)	97.39	19.735

and longest-feasible fitting fixed. WHOLE, half-, quarter-, and eighth-scale grids have 7/13/25/49 candidates. At 10 dB, QUARTER minus WHOLE gains $0.660\,[0.515, 0.814]$ dB, while EIGHTH adds $0.123\,[0.079, 0.169]$ dB over QUARTER. The denser grid also yields small PSNR/LPIPS gains at other SNRs; the main quarter grid remains the predefined configuration.

A receiver diagnostic on the same 100 sources separates direct boundary-residual correction from its propagation to finer scales. At 10 dB these contributions are respectively 0.220 and

TABLE II

PERCEPTUAL DISTANCE AND IMAGE-FEATURE SIMILARITY ON COMMON500. EACH CELL IS LPIPS-ALEX ↓ / CLIP-I ↑.

Method	4 dB	10 dB	19 dB
NeST-Com	0.263 / 0.821	0.171 / 0.873	0.118 / 0.900
Latent JSCC	0.219 / 0.741	0.173 / 0.789	0.158 / 0.804
SwinJSCC	0.530 / 0.627	0.469 / 0.669	0.440 / 0.697
BPG + LDPC	0.725 / 0.586	0.589 / 0.635	0.458 / 0.684
ARPC + LDPC	0.512 / 0.644	0.341 / 0.713	0.250 / 0.788
Ada (adapted)	0.395 / 0.828	0.167 / 0.884	0.138 / 0.892

0.440 dB. Propagation contributes zero at 19 dB, where the boundary is the last scale. These are mechanism diagnostics within the existing population.

C. Comparison With Existing Methods

Fig. 2 and Table II show a quality tradeoff. NeST-Com exceeds adapted Ada in PSNR by 4.116/0.455/1.226 dB at 4/10/19 dB, respectively. At 10 dB, Ada has lower LPIPS and DreamSim and higher CLIP-I, with source-paired CIs excluding zero. Ada also has higher CLIP-I at 4 dB and slightly lower DreamSim at 19 dB; NeST has better PSNR, LPIPS, and CLIP-I at 19 dB. Ada delivers all 256 tokens at 19 dB, so no tail sampling occurs; its approximately 58% source-capacity utilization reflects completion of its representation.

SwinJSCC and BPG have higher PSNR than NeST at every tested SNR. Latent JSCC has higher PSNR at 4/10 dB and lower LPIPS at 4 dB. Thus perceptual criteria and pixel error need not rank systems alike. NeST's PSNR advantage over ARPC is 3.26–3.97 dB. On matched100, replacing ARPC's original sampling by argmax or a predefined lower temperature changes mean PSNR by up to about 0.15 dB in the tested conditions.

Fig. 3 shows two selected development examples. Ada retains crisp pastry folds and dog fur; NeST and Latent JSCC smooth different details, while ARPC changes visible object contours and texture. For the dog, Latent JSCC has the lowest displayed LPIPS, whereas Ada has the lowest DreamSim. These individual outcomes complement the source-level comparison.

D. Additional Results and Processing Cost

On Kodak, all four paired metrics favor NeST over Ada at 4 dB. Only PSNR has a nonzero paired difference at 10 dB; at 19 dB, PSNR and LPIPS favor NeST while DreamSim/CLIP-I CIs cross zero.

The existing $N = 2048$ comparison uses 100 sources and two paid 1024-symbol subframes. Its 4/10 dB conditions are adaptive, whereas 19 dB delivers both complete representations. At 19 dB, NeST exceeds ARPC by 1.806 dB PSNR, but ARPC has better DISTS [17]: NeST minus ARPC is $+0.0054\,[+0.0024, +0.0086]$. Representation and decoding therefore remain relevant beyond payload fitting.

Measured software processing latency at 10 dB averages 619.71 ms/image for NeST, 42.48 for Latent JSCC, 44.69 for SwinJSCC, 1520.30 for ARPC, and 2293.01 for BPG (RTX PRO 6000, six CPU cores, batch one, 16 development images measured three times after warm-up). These totals sum TX/RX software processing; Ada has not been timed under this protocol.

IV. Conclusion

NeST-Com connects within-scale endpoints, terminated-length fitting, and prefix-constrained reconstruction under a fixed symbol budget. Shared-model evaluations show improved capacity utilization and reconstruction quality, while controlled grids reveal further gains from denser endpoints. Six-system results expose pixel–perceptual tradeoffs, including operating points favoring

adapted Ada. Calibration objectives and SNR mismatch remain useful directions for bounded evaluation.

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